UX Research, Testing & Design Systems

Purpose of UX Research, Testing & Design Systems

Great UI/UX design is **not based on assumptions**.
It is based on **user research, real testing, and consistent systems**.

At **ENGLORAY Tech Group**, this resource ensures that:

- Design decisions are data-backed
- User problems are identified early
- Interfaces are usable and scalable
- Products remain consistent as they grow

This resource connects **user needs → design decisions → scalable systems**.

1. UX Research (Understanding the User)

UX research helps designers **understand real users**, not imagined ones.

Research Methods Used

User Interviews

- One-on-one discussions with users
- Understand pain points, behavior, expectations

Surveys & Questionnaires

- Collect data from multiple users
- Identify patterns and trends

Competitive Analysis

- Study similar products
- Learn what works and what fails

- ***Persona Creation***

  - Create user profiles (age, goals, frustrations)
  - Design with a clear target user in mind

- ***Empathy Mapping***

  - Understand what users think, feel, say, and do

Outcome: Designers clearly understand *who* they are designing for.

2. Usability Testing (Validating the Design)

Testing ensures the design **actually works for users**.

Testing Methods Used

- Task-based usability testing
- Prototype testing
- A/B testing
- Feedback sessions

Tools Commonly Used

- **Maze**
- **Hotjar**
- **UserTesting**

Why Testing Matters

- Finds usability issues early
- Reduces rework and development cost
- Improves user satisfaction

Outcome: Designs are improved *before* development starts.

3. UX Metrics & Insights

Design success is measured, not guessed.

Metrics Used

- Task success rate
- Time on task
- Error rate
- User satisfaction

Why This Is Important

- Helps compare design versions
- Shows real improvement
- Supports design decisions with data

4. Design Systems (Consistency & Scalability)

Design systems are **a single source of truth** for UI design.

What a Design System Includes

- Color tokens
- Typography styles
- Buttons & UI components
- Spacing & grid rules
- Interaction guidelines

Standards Followed

- Material Design (Google)
- Apple Human Interface Guidelines
- IBM Carbon Design System

Why Design Systems Matter

- Consistent UI across screens
- Faster design & development
- Easy updates at scale

5. Accessibility & Inclusive Design

Designs must work for **all users**, including those with disabilities.

Accessibility Practices

- Proper color contrast
- Readable font sizes
- Keyboard navigation
- Screen reader compatibility

Standards Followed

- WCAG accessibility guidelines

Outcome: Products are inclusive and legally safe.

6. Iteration & Continuous Improvement

UX is an ongoing process.

ENGLORAY UX Cycle

Research → Design → Test → Improve → Repeat

Why This Matters

- User needs change

- Products evolve
- Continuous improvement keeps products relevant

Final Outcome of UX Research, Testing & Design Systems

With this resource, ENGLORAY UI/UX designers can:

✓ Understand real user needs
✓ Validate designs with testing
✓ Reduce usability issues
✓ Build scalable design systems
✓ Deliver consistent, user-friendly products